

Chapter 12

I/O Systems

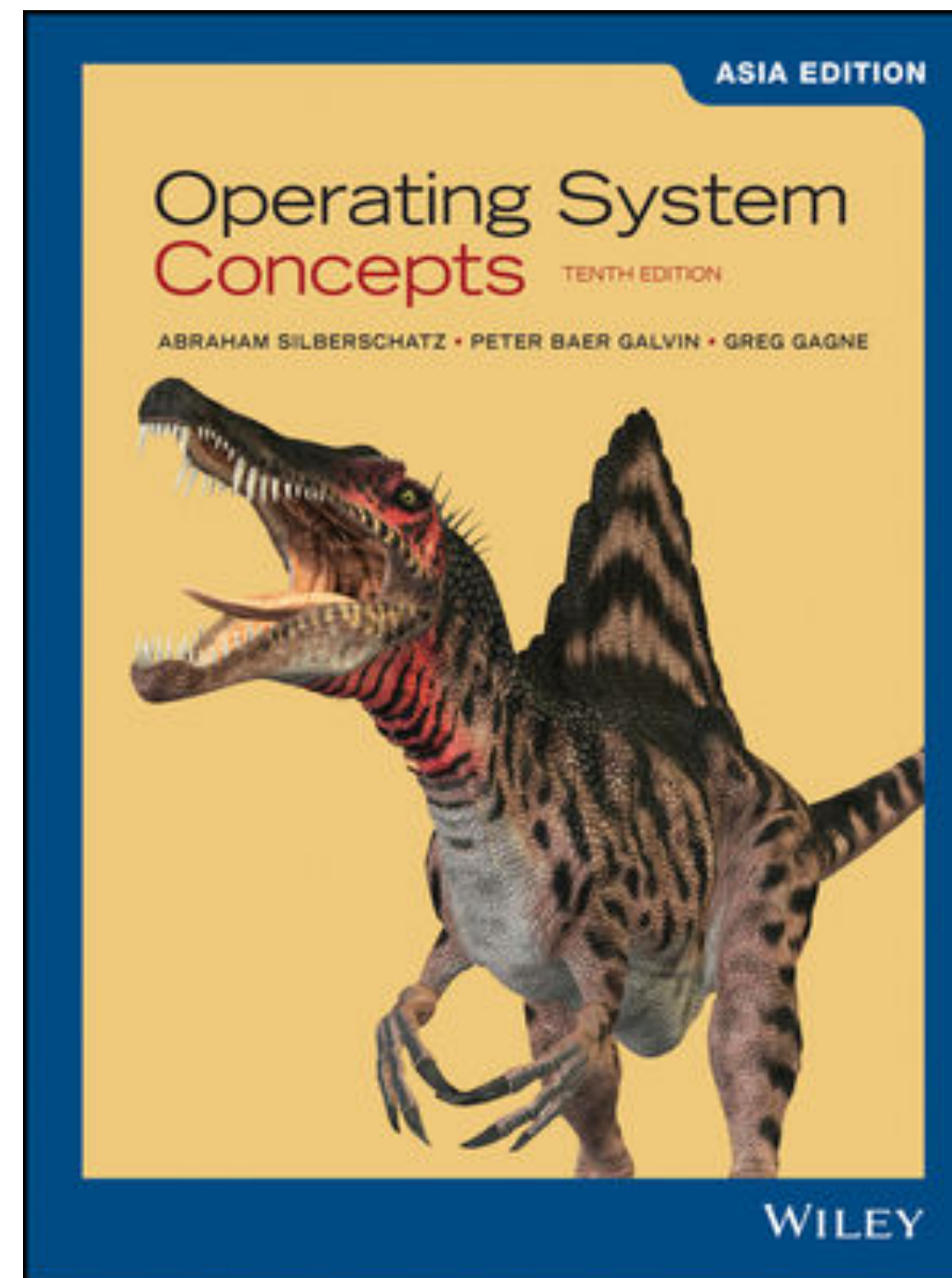
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Textbook

- Abraham Silberschatz, Peter B. Galvin, Greg Gagne, "Operating System Concepts, Asia Edition," 10/e, ISBN: 978-1-119-58616-6, Wiley



Memory-Mapped I/O

- Memory-Mapped I/O (MMIO) is a technique where I/O device **registers** or device **memory** are mapped into **the address space of the CPU**
 - allowing the CPU to interact with I/O devices **using regular memory access instructions** (such as load and store)
 - do not need to support **special I/O instructions** like IN and OUT
 - instead, all I/O operations are done through memory accesses
- Modern architectures such as ARM, RISC-V, and MIPS use Memory-Mapped I/O rather than **Port-Mapped I/O**

Polling / Interrupts

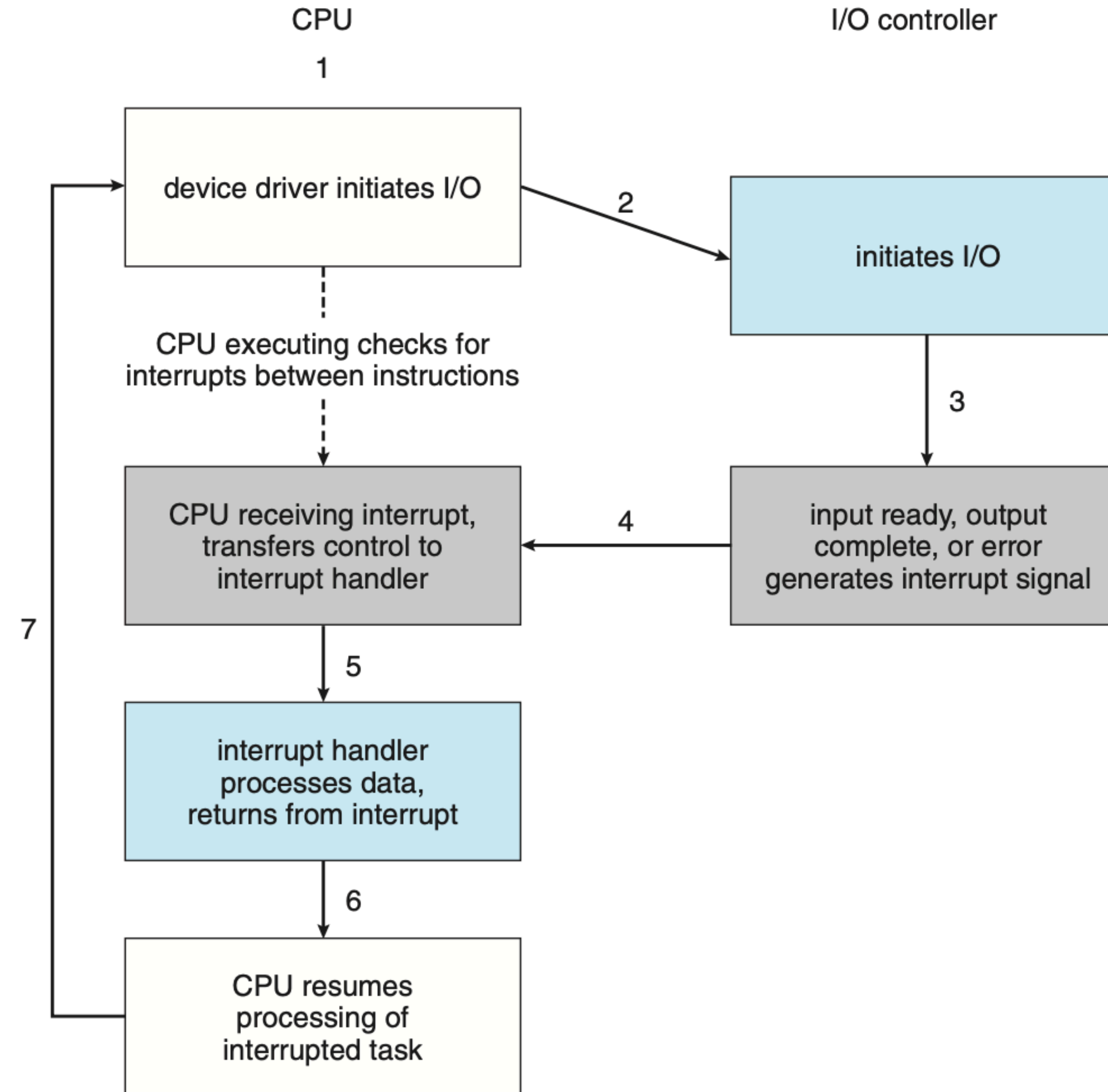
- Polling and Interrupts are two fundamental mechanisms used by the CPU to handle I/O operations with external devices
- Polling is a method where the CPU **repeatedly checks** (or “polls”) the status of an I/O device at regular intervals to determine whether it needs attention
- An interrupt is a signal sent by an I/O device to the CPU **asynchronously** to indicate that it needs attention.

Polling / Interrupts

Comparison: Polling vs Interrupt

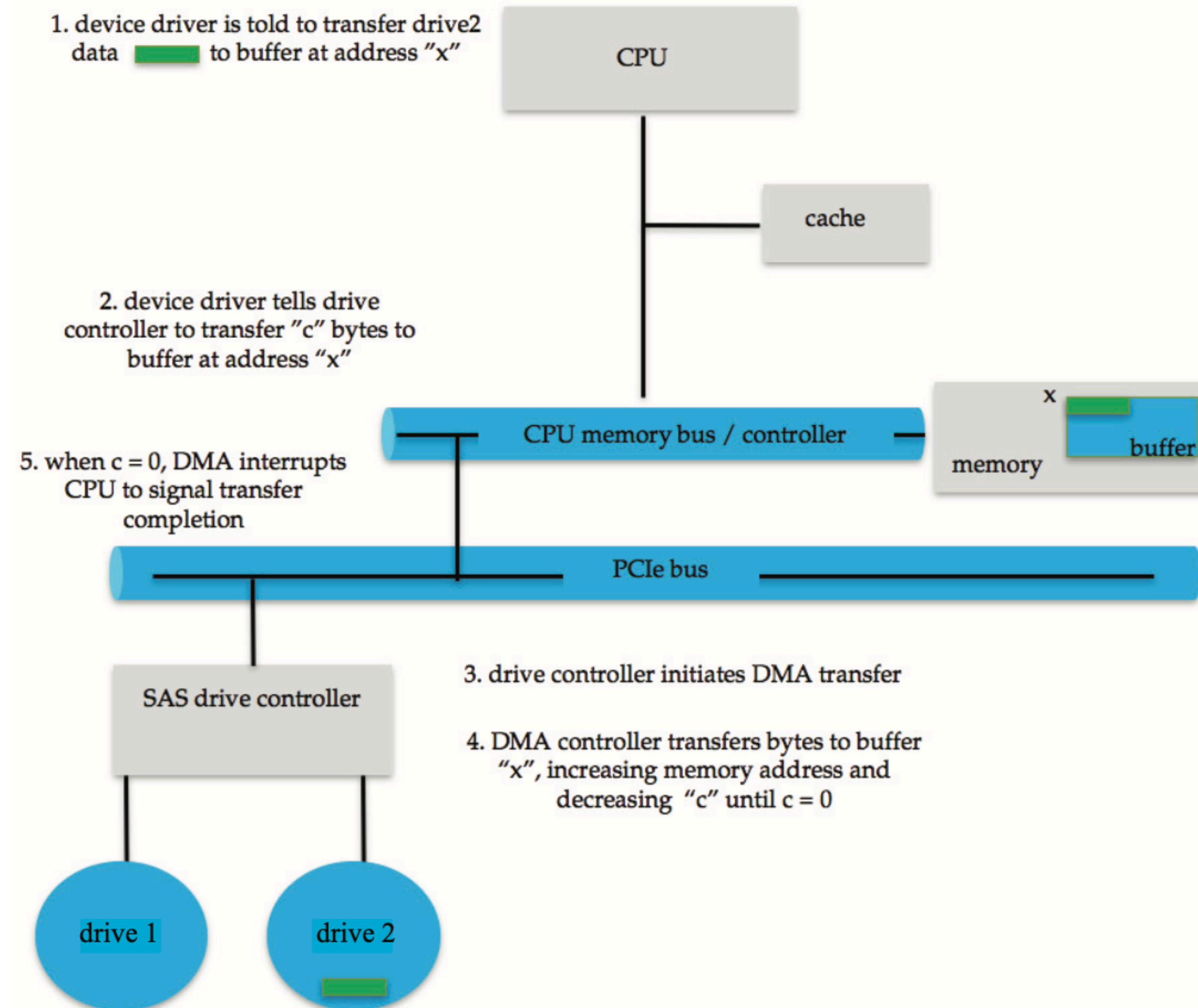
Feature	Polling	Interrupt
CPU involvement	Actively checks device	Passive until signaled by device
Efficiency	Low (busy-waiting)	High (event-driven)
Complexity	Simple	More complex
CPU usage	High	Low
Responsiveness	Delayed (depends on check interval)	Immediate
Suitable for	Simple, low-speed I/O	Real-time or high-speed I/O

Interrupt-driven I/O cycle

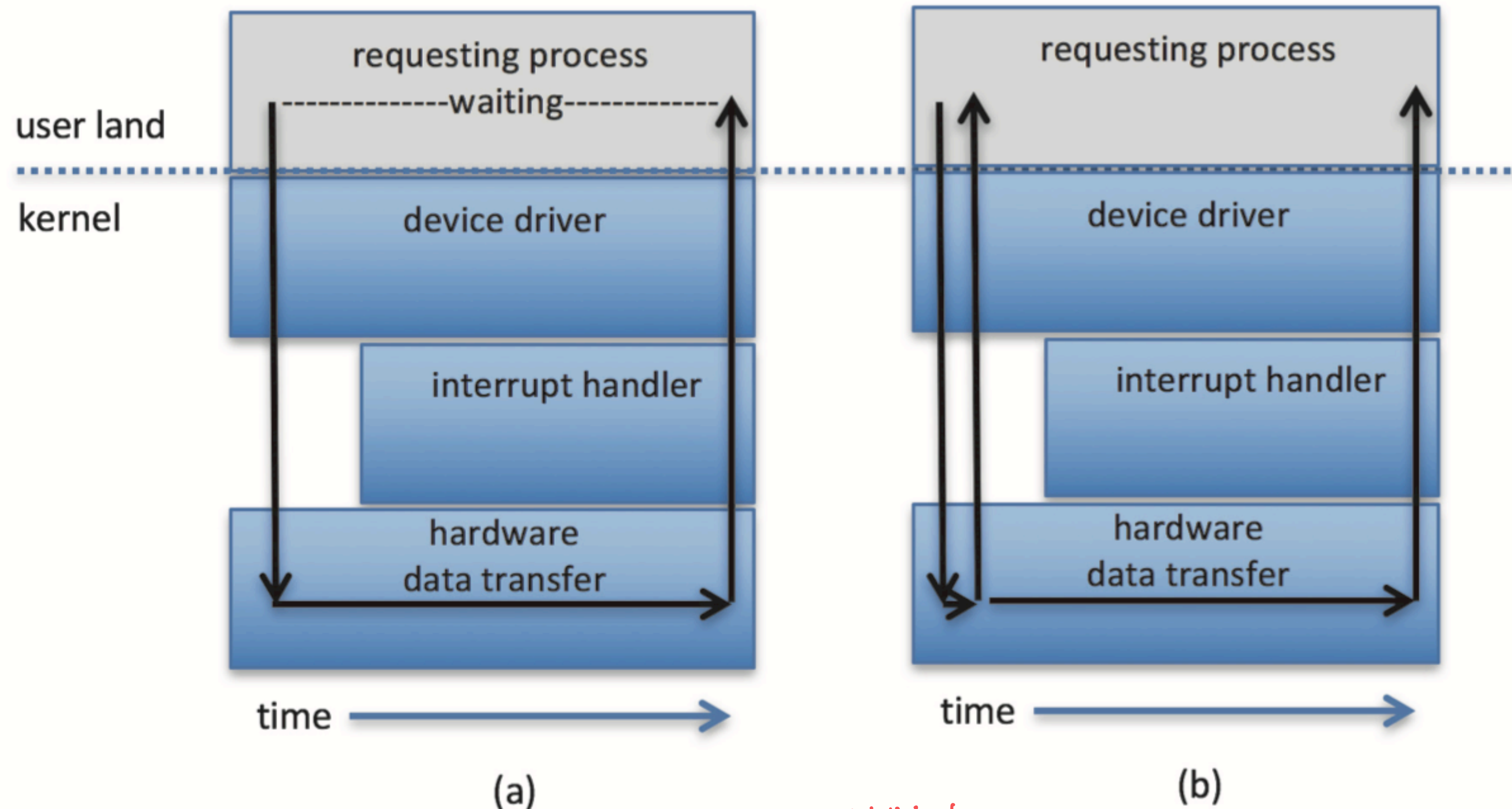


Direct Memory Access

- Used to avoid programmed I/O (one byte at a time) for large data movement



Synchronous / Asynchronous



I/O 時不能做其他事

Two I/O methods: (a) synchronous and (b) asynchronous.

I/O 時可以同時做其他事